Univalence implies weak function extensionality

The Univalent Foundations Program

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Editorial note. The counted statement is the labelled theorem ‘uatowfe‘ (source lines 957-961, Theorem 4.9.4); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the corollary that names the map alpha, invoked in the statement and needed for the label to be defined inside the package.

Editorial note. The retained counter values (chapter 4, section 9, theorem offset 2) make the retained context print as Theorem 4.9.3 and the counted theorem as Theorem 4.9.4, matching the original book numbering taken from the label-number dump.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: univalence implies weak function extensionality (Theorem 4.9.4).

Allowed standard prerequisites (editorial).

- Dependent product of contractible types. prd x:A P(x) is contractible when each P(x) is contractible.. It is the conclusion of the theorem.
- Retract. X is a retract of Y when there are maps in both directions whose composite is homotopic to the identity on X.. The proof exhibits prd x:A P(x) as a retract of the fiber of alpha at the identity, whence contractibility transfers.
- Uniqueness principle for dependent function types. A dependent function is equal to its own eta-expansion.. It identifies psi(varphi(f)) with f at source line 972.

Corollary 4.9.3. *Let $P : A \to \mathcal{U}$ be a family of contractible types, i.e. $\prod_{(x:A)} \mathsf{isContr}(P(x))$. Then the projection $\mathsf{pr}_1 : (\sum_{(x:A)} P(x)) \to A$ is an equivalence. Assuming $\mathcal{U}$ is univalent, it follows immediately that post-composition with pr_1 gives an equivalence*

$$\alpha : \Big(A \to \sum_{x:A} P(x)\Big) \simeq (A \to A).$$

Theorem 4.9.4. *In a univalent universe $\mathcal{U}$, suppose that $P : A \to \mathcal{U}$ is a family of contractible types and let α be the function of Corollary 4.9.3. Then $\prod_{(x:A)} P(x)$ is a retract of $\mathsf{fib}_\alpha(\mathsf{id}_A)$. As a consequence, $\prod_{(x:A)} P(x)$ is contractible. In other words, the univalence axiom implies the weak function extensionality principle.*

Proof. Define the functions

$$\begin{aligned} \varphi &: (\textstyle\prod_{(x:A)} P(x)) \to \mathsf{fib}_\alpha(\mathsf{id}_A), \\ \varphi(f) &:\equiv (\lambda x.\, (x, f(x)), \mathsf{refl}_{\mathsf{id}_A}), \end{aligned}$$

and

$$\begin{aligned} \psi &: \mathsf{fib}_\alpha(\mathsf{id}_A) \to \textstyle\prod_{(x:A)} P(x), \\ \psi(g, p) &:\equiv \lambda x.\, \mathsf{happly}(p, x)_*(\mathsf{pr}_2(g(x))). \end{aligned}$$

Then $\psi(\varphi(f)) = \lambda x.\, f(x)$, which is f, by the uniqueness principle for dependent function types. □